

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1923

Montpelier Electric Generating Station 100 kV
December 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1923. The Interconnection Customer proposes a 126.6 MW solar plus storage facility interconnecting at the Montpelier Electric Generating Station 100 kV substation in the Indiana area. The request was received on November 01, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 646 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$700,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STATE STREET / ROBISON PARK SUBSTATION 230kV	TREGO / SWAN VLY SUBSTATION 500kV	105.8	2,059	18.7
INTALCO / TAHITI 100kV	HARTFORD CITY / BEDFORD RURAL 100kV	102.8	109	11.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STATE ST / ROBISON PARK 230kV	Reconductor line	\$2,776,000	2% (\$41,000)
INTALCO / TAHITI 100kV	Reconductor line	\$3,800,000	17% (\$659,000)
	Total allocated to project		\$700,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.